

SMART INDIA HACKATHON (SIH)

Patient Case-Taking Software

Deep Functional Documentation - Approved Project Design

Problem Statement	Patient Case-Taking Software
Organization	Ministry of Ayush
Department	All India Institute of Ayurveda
Category	Software
Theme	MedTech / BioTech / HealthTech
Problem Statement ID	26047

Core Principle

The AI does not provide an autonomous final diagnosis. It collects and structures the patient's clinical history, digitizes supporting medical documents, generates a physician-ready report, and keeps the doctor in control of clinical diagnosis and treatment.

1. Project Overview

The proposed system is a patient-facing clinical intake platform designed to reduce the history-taking bottleneck in busy hospital OPDs. A patient can register, communicate with an AI consultation assistant through voice, optionally provide previous medical documents, review the generated patient report, and obtain a structured PDF that can be used by a physician during consultation.

The project is intentionally centered on clinical history acquisition rather than autonomous diagnosis. The software acts as a structured digital case-taking layer between the patient and the doctor. The extended design also includes patient-facing hospital services and a separate authenticated hospital administration portal.

1.1 Core Goals

- Reduce repetitive manual history taking before the patient meets the doctor.
- Capture patient complaints and history through a simple voice-led interaction.
- Allow previous prescriptions, lab reports, and discharge summaries to contribute to the final patient report.
- Generate a concise, structured and physician-readable clinical history report.
- Maintain patient report history for later viewing, downloading and printing.
- Provide Hindi and English interaction support with accessibility-oriented UI choices.
- Provide hospital-facing services without mixing them into the core AI diagnosis workflow.
- Keep final medical diagnosis and treatment decisions with the physician.

1.2 High-Level User Journey

Account Access -> Patient Registration -> AI Consultation -> Previous Report Upload (optional) -> AI Patient Report -> Review/Edit -> Confirm -> PDF Generation -> My Report History -> Hospital Services

1.3 Approved Module Map

Module	Name	Primary Purpose
1	Patient Registration	Capture essential patient profile information.
2	AI Consultation	Voice-led case taking and structured AI patient report generation.
3	My Report History	Access generated and stored patient reports.
4	Notification Center	Appointments, medicine reminders, report and hospital notifications.
5	My Profile & Settings	Manage account/profile information and preferences.
6	Multi-Language Support	Hindi/English interface and voice-language support.
7	Help & Support	Guidance, FAQs, support, feedback and issue reporting.
8	Hospital Services	Patient-facing hospital appointments, pharmacy, lab and service information.
9	Hospital Admin Portal	Authorized administrative management and CRUD operations.

2. Module 1 - Patient Registration

2.1 Objective

Patient Registration is the first clinical-profile module after account access. Its purpose is to collect only the essential patient information required to identify the patient and establish a usable hospital record without making the onboarding process unnecessarily long.

2.2 Design Principle

Registration must remain simple. Detailed clinical history is intentionally not collected here because the AI Consultation module will collect it conversationally. The registration form should therefore focus on stable demographic and baseline information.

2.3 Approved Registration Fields

#	Field	Status	Purpose
1	Full Name	Required	Primary patient identification.
2	Date of Birth	Required	Used to derive age and support clinical context.
3	Gender	Required	Basic demographic information.
4	Mobile Number	Required	Patient contact and account communication.
5	Email Address	Required	Digital communication and account reference.
6	Address	Required	Patient residential/contact information.
7	Blood Group	Optional	Captured only when known by the patient.
8	Height	Required	Baseline physical information.
9	Weight	Required	Baseline physical information.
10	Existing Diseases	Checkbox	Quick baseline selection such as diabetes, BP, asthma or none.

2.4 Existing Diseases Control

The Existing Diseases field should use a simple checkbox-based interaction rather than a long medical form. Example selections can include Diabetes, Hypertension/BP, Asthma and None. More detailed medical history is intentionally deferred to the AI consultation.

2.5 Registration Workflow

- Patient opens the Patient Registration module after account access.
- The system presents a clean form with the approved fields.
- The patient completes all required fields and optionally enters the blood group.
- The system validates required fields and input formats.
- The patient presses Submit.
- The patient profile is stored and becomes available to later modules.
- The user can later maintain profile-related information from My Profile & Settings.

2.6 UI Requirements

The page should use large readable labels, clear input fields, visible required/optional indicators, minimal visual clutter, and a prominent Submit button. Profile photo is not required in this registration form; it belongs to the profile area.

2.7 Validation and Edge Cases

- Prevent submission when mandatory fields are empty.
- Validate email and mobile-number formats.
- Avoid duplicate patient records when the same account already has a completed patient profile.
- Allow Blood Group to remain empty when unknown.
- Ensure the Existing Diseases control supports a clear None selection.

3. Module 2 - AI Consultation

3.1 Objective

AI Consultation is the core module of the project. It conducts a natural patient interview, captures the patient's current problem and relevant history, optionally incorporates previous medical documents, and converts the collected information into a structured patient report.

The assistant must dynamically ask follow-up questions based on what the patient says. It must not pretend to measure physical signs that were never provided, and it must not issue a final autonomous diagnosis.

3.2 Sub-Module 2.1 - Voice Assistant

When the patient selects AI Consultation, the Voice Assistant begins the clinical conversation. The assistant first welcomes the patient, confirms the preferred conversation language, and then asks an open question such as: **What problem are you facing today?** The patient - not the AI - describes the complaint in natural language.

Voice Conversation Flow

- Welcome greeting.
- Language confirmation.
- Open chief-complaint question.
- Patient explains the problem naturally.
- AI asks relevant follow-up questions based on the patient's own answer.
- The assistant captures details such as onset, duration, severity/context, related symptoms, medicines already taken and response to those medicines when relevant.
- The conversation continues until the history is sufficiently complete.
- The patient may optionally upload a previous medical report before final submission.
- The patient presses Submit to end the active conversation.

Previous Report Upload inside Voice Assistant

An optional **Upload Previous Report** control is available near the submission stage. The patient can provide previous prescriptions, lab reports or discharge summaries. The system should support printed and handwritten documents and multilingual medical content. OCR/document intelligence converts the document into usable clinical data before it contributes to the AI Patient Report.

Document Intelligence Requirements

- High-accuracy OCR for printed medical documents.
- Handwritten medical document reading where feasible.
- Multilingual document handling.
- Extraction of important clinical entities such as medicines, diagnoses mentioned in prior documents, investigation values and procedure/surgery history.
- Chronological use of prior information when dates are available.
- Previous document content should support the report; it should not silently become an autonomous diagnosis.

3.3 Sub-Module 2.2 - AI Patient Report

After the patient submits the conversation, the active voice interaction ends. The captured conversation and any uploaded previous documents are processed into a structured AI Patient Report.

Report Structure

- Patient identification and basic profile context.
- Chief complaint.
- History of present illness / current problem narrative.
- Relevant past history collected during the conversation.
- Medicines already taken and patient-reported response.
- Relevant allergies/drug history when captured.
- Relevant information extracted from uploaded previous documents.
- Concise patient report summary for rapid physician reading.

Review, Edit and Confirmation

Before finalization, the patient can **Review** the generated report and use an **Edit** option to correct wrongly captured information. After review, the patient presses **Confirm**. Confirmation locks the patient-facing final version for generation and produces the report as a PDF/document suitable for use during the physician consultation.

Physician Control

The report is a structured clinical-history draft, not a final diagnosis. The physician remains responsible for examination, clinical reasoning, diagnosis, treatment and prescription. This principle directly preserves the intended role of the software as a case-taking assistant.

3.4 Sub-Module 2.3 - AI Consultation Guide

The guide helps a first-time user understand how to interact with the AI consultation interface.

- Speak slowly and clearly.
- Describe your problem in your own words.
- Provide truthful and complete symptom information.
- Mention when the problem started and relevant medicines already taken.
- Use Upload Previous Report when an earlier prescription, lab report or discharge summary is available.
- Review the generated report before confirming it.
- Use Hindi or English according to the selected language preference.

3.5 End-to-End AI Consultation Flow

Start AI Consultation -> Welcome -> Language Confirmation -> Patient Narration -> Dynamic Follow-up Questions -> Optional Previous Report Upload -> Submit -> AI Patient Report -> Review/Edit -> Confirm -> PDF Generation -> Save to Report History

4. Module 3 - My Report History

My Report History stores and presents the patient's generated consultation reports and relevant stored medical records. It is an access-oriented module: the patient can view records but does not receive administrative delete permissions.

4.1 Patient Report Actions

- View a report.
- Review the content of a report.
- Download the report as PDF.
- Print the report.
- Share the report with the doctor/hospital when required.
- Search and filter stored reports.

4.2 Search and Filter

A Search/Filter control helps the patient locate older records. Approved filter concepts include date and report name/type. Because the user is already authenticated, patient-name filtering is not necessary for a personal report history page.

4.3 Record Integrity

Patient-facing delete operations should not be provided for formal medical records. Administrative record management belongs to the authenticated Hospital Admin Portal.

5. Module 4 - Notification Center

The Notification Center provides timely patient-facing updates connected to hospital activity and patient care.

5.1 Approved Notification Categories

Appointment Notifications: Appointment confirmation, timing changes and relevant appointment updates.

Medicine Reminders: Reminders for medicine timing and dose information when such information is available in the system.

Report Notifications: Alerts when an AI patient report or hospital/lab report becomes available.

Hospital Notifications: Important hospital announcements and service-related updates.

6. Module 5 - My Profile & Settings

This module manages the user's personal account/profile area. It is separate from the clinical case-taking flow.

6.1 Approved Areas

- Profile Information - view stored user/patient profile details.
- Edit Profile - update permitted profile information.
- Profile Photo - add or change the account image from the profile area.
- Account Settings - manage account-level preferences.
- Privacy & Permissions - manage relevant user permissions and consent-oriented settings.
- Notification Preferences - control available notification preferences.

7. Module 6 - Multi-Language Support

The approved initial language scope is Hindi and English. The module ensures that the software and the AI consultation experience can operate in the user's preferred language.

7.1 Approved Sub-Modules

- Language Selection.
- Voice Language Selection.
- Text Translation.
- Speech-to-Text Support.
- Language Preference Save.

7.2 Accessibility Intent

Language support should work with the wider accessibility design: large icons, readable controls, simple navigation and voice guidance. The design should remain usable for elderly and first-time users.

8. Module 7 - Help & Support

This module explains how to use the software and gives users a path to request help.

8.1 Approved Sub-Modules

- User Guide - how to use the website/software and its main features.
- Contact Support - hospital/software support contact route.
- FAQ - frequently asked questions and common usage answers.
- Feedback & Suggestions - user feedback and improvement suggestions.
- Report an Issue - submit software/service problems.
- Tutorial / Video Guide - optional visual guidance for first-time and elderly users.

9. Module 8 - Hospital Services

Hospital Services is the patient-facing service area. It expands the system beyond the AI case-taking flow while keeping hospital administration separate. The user can discover and access hospital services without receiving administrative privileges.

9.1 Doctor Appointment Booking

Patients can view doctor profiles, specialization, working days, availability/timings and appointment slots, and can book an appointment. The patient should also be able to see the booked appointment time and relevant status information. Doctor-related information should be presented clearly in Hindi and English where appropriate.

9.2 Hospital Pharmacy

The pharmacy area shows medicine availability and pharmacy service information. The patient can select eligible medicines for online purchase, choose hospital pickup or available delivery options, and view pharmacy operating information. Any prescription-controlled workflow should be handled according to hospital policy rather than bypassed by the interface.

9.3 Lab Test Booking

Patients can view available diagnostic/lab tests, book a test, select an available date/time, read preparation instructions, and later access the resulting report when the hospital makes it available.

9.4 Hospital Departments & Services

This section presents the services provided by the hospital so the patient can understand what is available before visiting. Examples may include OPD services, emergency care, doctor consultation, laboratory diagnostics, surgery-related services, ICU, blood bank and other hospital-specific facilities.

9.5 Doctor Directory

A searchable/list-based doctor directory provides doctor name, specialization, working days and hospital timings so users can check whether a relevant doctor is expected to be available.

9.6 Ambulance Request

The patient can access the hospital's ambulance request/contact pathway. Emergency-related information should remain quick to reach and should not force an emergency patient through the AI consultation workflow.

9.7 Hospital Contact & Location

The module provides hospital contact information and location details required by patients and attendants.

9.8 Emergency Care

Emergency care information is included as part of hospital services rather than creating an unnecessarily complex duplicate booking module. A patient in an emergency should be directed toward immediate hospital/emergency assistance, not required to first complete an AI-generated case report.

10. Module 9 - Hospital Admin Portal

The Hospital Admin Portal is a separate authenticated administrative area. It is not exposed as a normal patient feature. Authorized hospital administrators use it to manage operational data and perform permitted CRUD (Create, Read, Update, Delete) operations.

10.1 Security and Access Principle

Admin access must be role-based and authenticated. Patient users must never receive administrative CRUD permissions. Administrative actions should be auditable and limited to authorized hospital personnel.

10.2 Approved Administrative Areas

Admin Login & Authentication: Secure authenticated entry to the administrative portal.

Patient Management: Create, read, update and administratively manage patient records according to hospital policy.

Doctor Management: Add/update doctor profiles, specialization, schedules and availability.

Appointment Management: View, approve, reschedule or cancel appointment records where hospital workflow requires it.

Hospital Pharmacy Management: Manage medicine inventory, stock and availability.

Lab Test Management: Manage test offerings, bookings and report-related hospital workflows.

Departments & Services Management: Maintain the hospital's displayed departments and patient-facing services.

Emergency / Ambulance Request Management: View and manage requests routed to hospital operations.

Notifications & Announcements: Send relevant hospital announcements and patient-facing updates.

Reports & Basic Analytics: View operational summaries needed for administration.

10.3 CRUD Responsibility

CRUD is an administrative capability, not a patient capability. The portal should clearly separate data creation, viewing, updating and deletion actions and protect high-impact actions with suitable authorization and confirmation.

11. Cross-Cutting System Requirements

11.1 Homepage / Dashboard Direction

After account access, the software opens a clean dashboard. The header can include a search bar, home navigation, language control and user profile/avatar. The hospital's branding/visual identity may appear in the interface. Main modules should be presented as large, clear cards with an accessibility-oriented design.

11.2 Accessibility

- Large icons and clear labels.
- Simple navigation for elderly and first-time users.
- Voice guidance where useful.
- Hindi and English support.
- Avoid unnecessary form complexity.
- Keep emergency pathways quick and visible.

11.3 Privacy, Consent and ABDM Alignment

The original SIH problem statement expects consent-first handling of sensitive health data and alignment with the ABDM ecosystem. The implementation should therefore include appropriate consent, privacy and permission handling, and should be designed so structured health information can later integrate with hospital HIS/EMR and ABHA/FHIR workflows where available.

11.4 Clinical Safety Boundary

The system may structure patient-provided information, identify missing history questions, digitize documents and prepare a summary. It must not present its output as a substitute for physician examination or final diagnosis.

11.5 Data Flow Summary

User Account -> Patient Profile -> Voice History -> Optional Document OCR -> Structured Patient Report -> Patient Review -> Confirmed PDF -> Report History -> Physician/Hospital Use

12. SIH Requirement Alignment

SIH Requirement	Project Coverage
Patient-facing independent history capture	Covered by AI Consultation and Voice Assistant.
Natural voice + guided interaction	Covered by Voice Assistant and accessibility-oriented UI.
Adaptive clinical questioning	Covered by dynamic follow-up conversation flow.
Medical document digitization	Covered by optional Previous Report Upload and OCR/document intelligence.
Printed/handwritten multilingual OCR	Defined as an AI document-processing requirement.
Structured physician-ready summary	Covered by AI Patient Report and Patient Report Summary.
Editable/verifiable report	Covered by Review, Edit and Confirm flow.
No autonomous diagnosis	Explicitly enforced as a core design principle.
Hindi/English support	Covered by Multi-Language Support.
Privacy/consent orientation	Covered in Profile/Settings and cross-cutting requirements.
HIS/ABHA integration direction	Preserved as an implementation/integration requirement.

12.1 Scope Note

Hospital Services and the Hospital Admin Portal are extensions beyond the narrowest case-taking core. They are intentionally separated from the AI Consultation module so the central SIH requirement remains clear and demonstrable.

13. Implementation Planning Notes

Technology selection is intentionally deferred until the functional modules and flows are finalized. The next implementation phase can map each approved feature to frontend, backend, database, speech, OCR, AI, security and integration technologies.

13.1 Suggested Build Order

- Build account access, dashboard and Patient Registration UI.
- Build AI Consultation UI and conversation flow.
- Add previous-report upload and document-processing pipeline.
- Build AI Patient Report, review/edit/confirm and PDF output.
- Build My Report History.
- Add Notification Center, Profile & Settings and Multi-Language Support.
- Add Help & Support.
- Build patient-facing Hospital Services.
- Build the separate authenticated Hospital Admin Portal.
- Add integrations, security hardening, testing and deployment.

14. Final Project Definition

The SIH Patient Case-Taking Software is designed as an AI-assisted clinical intake platform in which a patient provides basic registration details, communicates naturally with a voice assistant, optionally supplies previous medical documents, and receives a structured patient-history report for physician use. The patient can retain and access report history, while hospital-facing service functions and administrative CRUD operations remain clearly separated into dedicated modules.

The project's defining value is not autonomous diagnosis. Its value is faster, more complete and more structured clinical history acquisition before the physician consultation.

END OF FUNCTIONAL DOCUMENTATION